Reading Consciousness from the Schematic

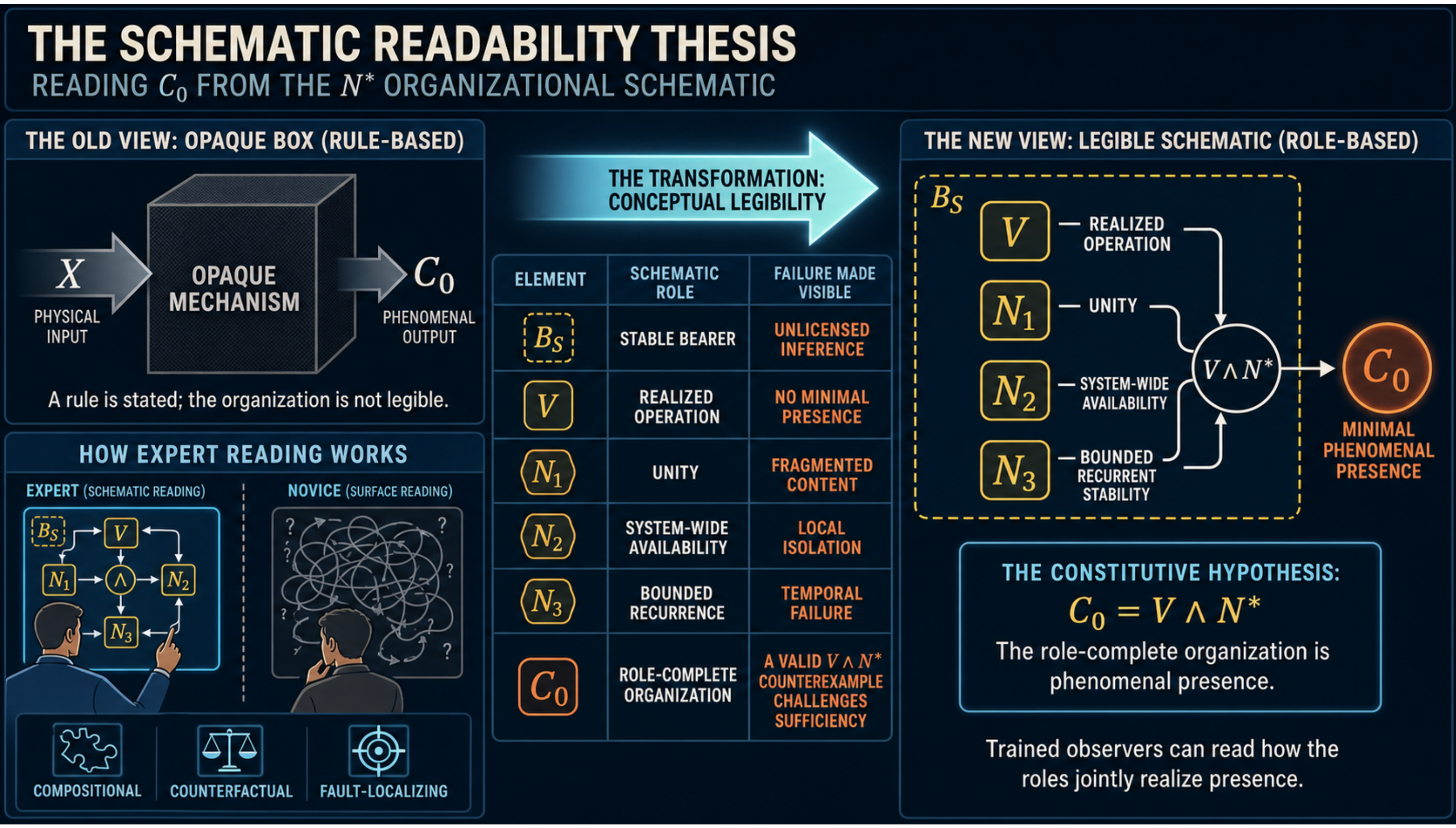
Structural Legibility in the N^* Model

Phil Stilwell

Independent Scholar
ORCID: 0009-0009-3725-8682

July 16, 2026

A visual preview:



Abstract

The N^* model proposes that minimal phenomenal presence, C_0, obtains exactly when an independently individuated and viable candidate system realizes integration or synergy (N_1), report-independent system-wide availability (N_2), and recurrent stability (N_3). Even if this biconditional is granted, a lacuna appears to remain: is the necessity of consciousness only a black-box rule attached to the dynamics, or can a competent observer understand why the organization is the phenomenal state rather than a merely correlated producer of it? This paper develops the latter possibility through the analogy of an electrical schematic. An engineer can inspect a diagram and infer a lamp's illumination or a motor's operation because the symbols have learned component semantics, their organization supports a state-by-state derivation, and alterations have localized counterfactual consequences. The output is not visually contained in the ink, and the diagram alone does not validate the component laws; nevertheless, the outcome is structurally legible. The paper argues that N^* has the same explanatory form when read under its constitutive hypothesis. System individuation supplies a bearer, N_1 supplies unity, N_2 supplies system-level causal presence, N_3 supplies bounded temporal presence, and V supplies realized operation. A trained reader can therefore pass through an explicit role vector rather than an uninterpreted lookup from a score to a phenomenal label. The proposal defines schematic legibility through compositional readout, counterfactual fault localization, and evidential independence; distinguishes intelligibility from proof; and answers objections involving circularity, zombies, functional-role substitution, and the disanalogy between mature electrical theory and consciousness science. The conclusion is conditional but substantive: if $C_0 = V \wedge N^*$ is true as a constitutive identity, consciousness is not merely guaranteed somewhere behind the model. It is readable from the organization in the way an operating state is readable from a well-understood schematic.

Keywords: phenomenal consciousness; phenomenal presence; N^* model; explanatory gap; mechanistic explanation; schematic representation; structural intelligibility; constitutive identity; functionalism; qualia

1. The Lacuna: Necessity Without Intelligibility

The N^* research program advances a deliberately strong hypothesis about minimal phenomenal presence (Stilwell, 2026a):

$$C_0(S, c; \mathscr{R}, \Delta) \iff V(S; \mathscr{R}, \Delta) \wedge N_1(S, c) \wedge N_2(S, c) \wedge N_3(S, c).$$

Annotation. S is a boundary-qualified candidate system; c is a content or content-bearing state; $\mathscr{R}$ is the operating regime; Δ is the evaluation interval; V is independent viability; N_1 is integration or synergy; N_2 is counterfactual system-wide availability; N_3 is recurrent stability; and C_0 is the fact that there is something it is like for S to undergo the state. The biconditional is an empirical and constitutive hypothesis, not a label generated from the right-hand side. For readability, the role terms suppress the frozen specification Θ; each is evaluated for the same boundary, regime, grain, interval, interventions, thresholds, and uncertainty policy.

The target is phenomenal rather than access consciousness: the minimal what-it-is-like fact, not reportability or flexible control (Nagel, 1974; Block, 1995).

The companion papers make the hypothesis progressively more disciplined. They require the candidate system to be individuated before consciousness is measured, specify availability without reducing it to report, preserve indeterminate and unlicensed outcomes, map phenomenal character only after the presence gate is passed, and audit each conjunct against rivals that omit it (Stilwell, 2026b, 2026c, 2026d, 2026e, 2026f). Those steps address scope, measurement, evidence, extension, and component necessity.

A further explanatory question remains. Suppose the biconditional survives. Would the theory tell us only that consciousness necessarily accompanies $V \wedge N^*$, as if an opaque box converted a dynamical input into a phenomenal output? Or could someone trained in the theory inspect the organization and understand why phenomenal presence is the state predicted there?

The difference can be stated as follows:

$$\text{black-box necessity:} \qquad X \Rightarrow C_0,$$

$$\text{structurally legible necessity:} \qquad X \Rightarrow \rho \Rightarrow C_0,$$

where ρ is an interpretable role profile connecting the organization to the minimal structure of phenomenal presence. In the first case, the reader knows only a rule. In the second, the reader can follow how the parts jointly realize a bearer, unity, system-level presence, temporal presence, and actual operation. The second relation does not make the hypothesis self-proving. It makes its content inspectable.

This paper argues that the N^* hypothesis is intended to support the second form. Its central thesis is:

> **Schematic-Legibility Thesis.** If $C_0 = V \wedge N^*$ is true as a constitutive identity, then a role-complete N^* organization makes phenomenal presence conceptually readable to a competent observer. The observer does not infer consciousness from an uninterpreted score. The observer reads the candidate bearer, unity, system-level availability, temporal maintenance, and realized operation from the organization, and recognizes their conjunction as the physical form of minimal phenomenal presence.

The thesis is conditional. It does not claim that a drawing proves consciousness or that intuitive vividness supplies evidence for the biconditional. Its claim concerns explanatory form: the proposed necessity is decomposable, counterfactual, and available to trained understanding.

2. What an Expert Reads in a Schematic

Consider an electrical schematic containing a source, conductors, a closed switch, a control element, and a lamp or motor. A novice may see standardized marks connected by lines. An engineer sees possible differences, current paths, loads, operating ranges, and dependencies. Given the input conditions and component specifications, the engineer can infer whether the lamp illuminates, the motor turns, or a protection device opens.

The conclusion is not available from the geometry of the ink alone. The reader must know what the symbols denote, how the components behave, whether idealizations are appropriate, and whether the represented circuit is actually powered. A diagram with an exhausted source, an underrated conductor, or an omitted ground path may support a different conclusion from one with the same superficial topology. Expert readability therefore depends on a structured package:

$$\mathcal{D}_E + \mathcal{K}_E + \mathcal{I}_E \vdash O_E.$$

Annotation. $\mathcal{D}_E$ is the electrical diagram, $\mathcal{K}_E$ is knowledge of component laws and representational conventions, $\mathcal{I}_E$ is the declared input and operating condition, and O_E is the inferred operating state. The turnstile marks a licensed derivation, not logical entailment from ink marks considered without semantics.

Three features make this inference explanatory rather than merely associative.

First, it is **compositional**. The reader can move from the state of each relevant component to the state of the organized circuit. Second, it is **counterfactual**. Opening the switch, breaking a conductor, changing the load, or removing the source produces a predictable difference. Third, it is **fault-localizing**. If the motor does not turn, the schematic constrains where the failure could lie and which observations would discriminate among alternatives.

Mechanistic explanation often has this form. It renders a phenomenon intelligible by identifying organized entities, activities, and dependencies whose coordinated operation produces or constitutes the target state (Machamer et al., 2000; Craver, 2007). A good schematic is valuable because it compresses those dependencies without hiding them. It permits the trained reader to mentally run the system.

The analogy is not that a conscious system is literally an electrical circuit, nor that each N^* conjunct corresponds to one electrical component. The analogy concerns the epistemic achievement. In both cases, an expert-readable representation can make a necessary operating state available through learned semantics, organization, and counterfactual reasoning.

3. The N^* Schematic

Let the complete phenomenal-presence schematic be represented as

$$\mathfrak{S}_{N^*}(S, c; \Theta) = \langle B_S, V, N_1, N_2, N_3 \rangle,$$

where B_S records that S passed an independent boundary criterion and Θ freezes the grain, horizon, regime, interventions, thresholds, and uncertainty policy. B_S is not an additional phenomenal conjunct. It licenses the referent to which all the other terms apply.

The schematic has two readings. A merely classificatory reading asks whether every gate equals one. A constitutive reading asks what organization those gates jointly describe.

Element	Schematic reading	Failure made visible
B_S	There is a sufficiently autonomous, inclusion-minimal candidate that could bear one system-level episode.	The measures lack a stable bearer, or multiple candidates remain unresolved.
V	The candidate is actually operating in a regime capable of realizing the relevant dynamics.	The diagram is inert, underpowered, disrupted, or outside the measures' validity envelope.

Element	Schematic reading	Failure made visible
N_1	The content is realized as an irreducibly joint state rather than a heap of independent local events.	The frozen candidate no longer supplies one unified content; smaller or fragmented candidates may need separate evaluation.
N_2	The content is causally available across functionally diverse recipients of the same candidate without requiring report.	The content remains an isolated local event, passive diffusion, or an output-specific artifact.
N_3	The content is maintained through bounded recurrent dynamics long enough to constitute a temporally present state.	The event is an evanescent feedforward transition or an unbounded pathological fixation.
C_0	The role-complete organization is, under the constitutive hypothesis, minimal phenomenal presence for S.	A valid $V \wedge N^*$ case with independent evidence against presence challenges sufficiency.

The table is not a verbal redescription pasted onto otherwise unrelated measures. Each role has an independent operational program and a characteristic failure mode. Integration must exceed reducible aggregation. Availability must survive report-channel removal and diffusion controls. Recurrence must be content-relevant and bounded. Viability must be assessed before the consciousness verdict. The boundary must not be chosen to maximize the eventual score.

The resulting intermediate profile is

$$\rho_{C_0} = \langle \text{bearer}, \text{realized operation}, \text{unity}, \text{system-level presence}, \text{temporal presence} \rangle.$$

A trained reader of $\mathfrak{S}_{N^*}$ does not jump directly from a Boolean vector to a phenomenal label. The reader reconstructs ρ_{C_0}. Under the identity hypothesis, this is not a list of effects consciousness happens to cause. It is the proposed physical-role decomposition of minimal presence itself.

3.1 The bearer is not supplied by a pronoun

Descriptions of consciousness often use phrases such as "available to the system" or "present for the subject" while leaving the referent unspecified. The boundary paper blocks that shortcut. The candidate must be generated from consciousness-independent evidence of dynamical autonomy and then frozen before N^* is evaluated (Stilwell, 2026c).

This prior individuation is part of what makes the schematic readable. An engineer cannot infer a circuit state without knowing which elements compose the circuit and which are inputs, loads, supports, or environmental conditions. Likewise, an N^* reader cannot infer a system-level state while moving the system boundary separately for integration, availability, and recurrence. B_S fixes the entity whose organization is being mentally run.

3.2 Unity, system-level presence, and temporal presence

The three N^* conjuncts answer different structural questions.

N_1 answers whether the state is one causally integrated content for the candidate rather than a coincident collection. N_2 answers whether that content occupies a system-level causal role rather than remaining locally encapsulated. N_3 answers whether that role is maintained through a bounded interval rather than occurring as an instantaneous transition. Together they describe not mere complexity but a unified content that is present across and through the operation of one candidate system.

The word "presence" in these glosses must not do illicit work. System-level presence is operationalized through counterfactual causal availability; temporal presence is operationalized through recurrent dwell. Neither definition contains a report of phenomenality. Their importance is structural: they realize roles that correspond to indispensable aspects of the minimal phenomenal target. The biconditional then makes the substantive identity claim that a viable system instantiating the complete role profile is phenomenally present.

3.3 Character is a further readout

The phenomenal-character extension adds a second schematic layer. Once the presence gate is licensed, a content-specific causal-trajectory geometry G_c may be mapped to an independently anchored quality space through a validated bridge Q_Φ (Stilwell, 2026e):

$$V \wedge N^* \quad \text{licenses presence,}$$

$$Q_\Phi(G_c, \Delta) \quad \text{differentiates character.}$$

This division matters for schematic reading. The N^* layer permits the inference that an episode is present. The geometry layer constrains which phenomenal relations the episode instantiates: similarity, intensity, valence, neighborhood, transition, and temporal path. A change in G_c can therefore alter the indicated experience without opening the presence circuit, so to speak. Conversely, rich geometry outside a licensed presence gate remains a structured representation without an ordinary phenomenal attribution.

4. A Criterion for Schematic Legibility

The phrase "a trained mind can see it" risks sounding private or rhetorical. It can be made more exact. Let M be a structured model, o a target state, and K a publicly teachable body of representational and component knowledge. Define schematic legibility as follows:

$$\text{Legible}(M, o \mid K) \iff D \wedge C \wedge F \wedge E.$$

The four conditions are:

1. **Decomposable semantics (D).** The relevant elements of M have stable role meanings that can be stated without simply repeating o.

2. **Compositional readout (C).** A competent reader can derive an intermediate organization profile from those elements and infer o through an explicit rule.
3. **Counterfactual fault localization (F).** Licensed changes to elements of M support differentiated predictions about which aspect of the profile fails or changes.
4. **Evidential independence (E).** The readout procedure does not count as evidence that its own constitutive rule is true; validation must come from observations, anchors, interventions, and rival comparison not generated by the readout.

For N^*, the readout has the form

$$\mathfrak{S}_{N^*} \Longrightarrow \rho_{C_0} \Longrightarrow C_0.$$

K_{N^*} contains the public meanings, measurement policies, and causal roles of the schematic elements. H_{C_0} is the constitutive hypothesis identifying the role-complete organization with phenomenal presence. The first arrow is mechanistic interpretation. The second is the theory's exposed psychophysical identity claim.

This separation prevents two opposite mistakes. One mistake says that because H_{C_0} is needed, everything before it is explanatorily idle. That ignores the decomposable role profile and its contrastive consequences. The other says that because the role profile is intelligible, H_{C_0} has been proven. That confuses understanding a hypothesis with validating it.

4.1 The counterfactual readout

The schematic earns transparency by supporting more than one positive verdict. It must also organize nearby failures:

- If B_S fails, the inference is unlicensed because no stable candidate bearer has been identified.
- If V fails, the represented organization is not operating in the regime to which the component semantics apply.
- If N_1 selectively fails while the candidate and remaining conditions are preserved, the full model withdraws the claim of one unified content for that candidate.
- If N_2 selectively fails, the content remains causally isolated from the candidate's functionally diverse organization.
- If N_3 selectively fails, the content does not achieve the bounded temporal maintenance required by the model.
- If G_c changes while the presence gate remains licensed, the predicted phenomenal character may change without loss of phenomenal presence.

These are not interchangeable negatives. They are a fault map. The ablation paper's diagnostic-set requirement provides the empirical counterpart: component necessity is tested only where the full model and a leave-one-out rival actually disagree while the candidate, regime, and retained conjuncts remain stable (Stilwell, 2026f).

4.2 What training consists in

The trained reader requires no special faculty of phenomenal intuition. Training consists in learning to:

1. hold the candidate boundary, grain, regime, and interval fixed;
2. distinguish causal integration from coactivation;
3. distinguish counterfactual availability from report or diffusion;
4. distinguish recurrent content maintenance from mere persistence;
5. apply the viability and uncertainty rules;
6. reconstruct the role profile; and
7. apply the constitutive hypothesis while keeping its evidential status explicit.

Two trained readers given the same frozen specification should therefore agree about the model's readout even if they disagree about whether the identity hypothesis is ultimately true. Schematic visibility is intersubjective competence, not private revelation.

5. From Correlation to Constitutive Readability

The schematic analogy clarifies three increasingly strong theoretical positions.

Position	Claim	What the reader can infer
Correlational marker	N^* covaries with evidence associated with consciousness.	A positive score raises confidence, but the internal organization may remain explanatorily incidental.
Black-box law	$V \wedge N^*$ reliably guarantees C_0.	Consciousness is necessary given the law, but the transition from dynamics to presence is opaque.
Constitutive schematic	C_0 is the role-complete $V \wedge N^*$ organization of a boundary-qualified system.	The reader reconstructs the minimal phenomenal profile and can localize how changes alter or defeat it.

The third position is the intended strong reading. It treats $C_0 = V \wedge N^*$ as a candidate theoretical identity discovered and tested a posteriori, not as a dictionary definition (Kripke, 1980). If true, the identity is stronger than regular co-occurrence. A system could not retain the complete constitutive organization while lacking the state that organization constitutes.

This is a metaphysical identity claim, not an assertion that phenomenal vocabulary is analytically derivable from nonphenomenal vocabulary. Schematic legibility concerns trained recognition under an empirically licensed identity.

This resembles other expert inferences only at the level of explanatory form. A circuit's operating state becomes necessary relative to component laws and boundary conditions. A molecule's behavior becomes intelligible relative to a structural and dynamical model. In each case, the observer learns to see a macrostate in an organized base rather than waiting for an extra event to be added. The N^* proposal says that minimal phenomenal presence should become similarly recognizable in a particular causal organization.

The analogy does not eliminate the explanatory-gap challenge identified by Levine (1983) and Chalmers (1995). A critic may grant every functional role and deny that the phenomenal conclusion follows. The present reply is not that the critic has made a formal mistake. It is that, under the identity hypothesis, the critic is treating a proposed constitution as if it were an external cause. Asking what additional event makes N^* produce consciousness would then resemble asking what further circuit event, beyond the operating state of the source, path, and load under their laws, makes that very operating state occur.

Whether that identity should be accepted remains empirical and philosophical work. The schematic thesis shows what acceptance would amount to: not faith in a hidden bridge, but acquisition of a stable capacity to read phenomenal presence through its organized physical roles.

6. What the Analogy Does Not License

The analogy is useful only if its limits remain visible.

First, **a representation is not a realization**. A printed circuit diagram does not illuminate a room, and a graph of N^* does not experience anything. The candidate system must actually realize the dynamics under V at the declared grain and horizon.

Second, **readability is not confirmation**. An elegant but false schematic can be perfectly intelligible. Independent anchors, near-miss cases, selective perturbations, transport tests, and comparisons with ablated rivals must determine whether the C_0 identity earns acceptance.

Third, **component semantics are theory-laden**. Engineers inherit mature component models supported by extensive measurement. N_1, N_2, and N_3 remain research constructs with competing estimators. Surrogate conflict must produce indeterminacy rather than a selectively favorable reading (Stilwell, 2026b).

Fourth, **legibility is indexed to a specification**. A conclusion may change with a defensible change in boundary, grain, interval, or operating regime. Such sensitivity limits the scope of the readout; it does not license choosing the specification after the desired phenomenal conclusion is known.

Fifth, **presence and character remain distinct**. Reading C_0 from the presence schematic does not reveal absolute qualitative labels. The geometry extension may constrain phenomenal relations while leaving global inversions or other automorphisms unresolved.

These limits keep the schematic proposal from becoming a visual version of stipulation.

7. Objections and Replies

Several objections trade on an ambiguity in "observing consciousness." Let A be an observer, S another boundary-qualified system, and q_S the token phenomenal character of one episode in S. Three epistemic relations must be distinguished:

1. A observes that S instantiates the organization constitutive of C_0;
2. A infers features of q_S from reports, behavior, interventions, and a licensed character map; and
3. A undergoes q_S from S's first-person position.

The first is the central observational ambition of the presence model. The second is the more limited ambition of the phenomenal-character extension. The third is neither required by the theory nor a coherent general standard for third-person science. It asks an observer not merely to detect another episode but to become acquainted with it as the subject whose episode it is.

Objection 1: The schematic contains consciousness only because the theory puts it there

The constitutive hypothesis is indeed required. No arrangement of uninterpreted marks entails phenomenality. But this is also true of ordinary schematics: without component semantics, a lamp symbol is only a mark. The relevant question is whether the semantics merely append an output label or expose an organized role profile. N^* does the latter by separating bearer, viability, unity, availability, and temporal maintenance, each with independent tests and failure modes.

The identity hypothesis is not thereby proven. It is made nonopaque. If the identity is independently supported, observing an active system realize the complete schematic is not observing a nonphenomenal mechanism plus an invisible phenomenal addition. It is observing the phenomenal episode under its causal-dynamical description. The schematic does not insert an unobserved output after the physical story; it identifies which organized physical state is the episode.

Objection 2: Availability and unity smuggle phenomenality into functional language

N_1 and N_2 are operationalized causally without using a consciousness verdict. Synergy, intervention-sensitive reach, functional diversity, latency, and route dependence can be tested in systems whose consciousness status is disputed. The phenomenal conclusion arises only from the full hypothesis. The structural gloss explains why those nonphenomenal roles are relevant; it does not turn them into hidden reports of experience.

Objection 3: A physical duplicate without experience remains conceivable

The zombie intuition challenges the truth or necessity of the identity, not the internal legibility of the proposal. A reader can understand exactly what N^* predicts and still reject it. If a genuine viable N^* duplicate without C_0 could be independently established, the sufficiency direction would fail.

But the imagined duplicate cannot be established merely by repeating that no observer has first-person access to it. That absence of access holds for ordinary conscious others as well. If the identity is true, observing the full realized N^* organization is observing C_0, although it is not undergoing the candidate's experience. The alleged zombie then removes a state while stipulating that its complete constitutive realization remains. The dispute is about whether the organization is constitutive, not about whether the observer can privately sample the candidate's phenomenology.

Objection 4: Consciousness cannot be observed from outside

This objection equivocates between observing that an episode exists and undergoing the episode. Suppose measurements establish that S is active, boundary-qualified, viable, integrated, system-wide available, and recurrently stable under a frozen specification. If H_{C_0} is true and independently licensed, those measurements disclose the state constitutive of phenomenal presence. The observer can therefore observe that phenomenal experience is occurring in S by reading its realized organization, just as an engineer observes that a powered circuit is in the state that illuminates its lamp. The phenomenology is not a second event hidden behind the measured organization.

What the observer does not acquire is S's first-person acquaintance with the episode. Formally,

$$\text{Observe}_A[C_0(S)] \nRightarrow \text{Undergo}_A[q_S].$$

This implication fails without making the observation defective. Scientific observation routinely identifies what state a system is in without reproducing that state in the observer. Here the difference is especially important because first-person acquaintance is indexed to the bearer. The observer sees the occurrence under a causal-dynamical mode of presentation; S undergoes the same occurrence under its phenomenal mode of presentation.

The epistemic asymmetry therefore concerns modes of access, not whether anything observable warrants the claim that phenomenology is occurring. Before the identity is well supported, N^* measurements are candidate evidence for consciousness. After it is independently supported, observation of the realized organization is observation of consciousness under the theory's public description.

Objection 5: Without access to the actual experience, the attribution cannot be verified

The demand for "the actual experience" can mean two things. It may mean sufficiently strong evidence about whether S experiences and how its experience is structured. That is a legitimate demand, answered imperfectly by calibrated report, behavior, causal perturbation, convergent anchors, no-report cases, and the quality-space bridge. The result may still be indeterminate or unlicensed.

Or it may mean that A must literally experience S's token episode as S experiences it. That demand is incoherent while A and S are stipulated to remain numerically distinct experiencers. If A undergoes a merely qualitatively identical episode, it remains A's episode rather than S's. If the episode is numerically the same token with the same first-person bearer, then the assumed boundary between two independent subjects has changed: the case is no longer one experiencer observing the private episode of another. It has become a problem of shared, overlapping, or re-individuated subjecthood.

Thus, no possible experiment could satisfy the strongest version of the demand while preserving the situation it was meant to test. Requiring token-identical first-person acquaintance would make knowledge of every other mind impossible by definition. The appropriate scientific requirement is not that one experiencer become another, but that public evidence reliably identify when another system instantiates the organization constitutive of experience and constrain what can responsibly be inferred about its character.

Objection 6: Experts often misread schematics

Yes. Readability is not infallibility. A schematic may omit a parasitic path, use a component outside its range, or idealize away a decisive condition. The corresponding response is to state the boundary, validity envelope, tolerances, and uncertainty rule. A theory becomes more trustworthy by making its possible misreadings diagnosable.

Objection 7: Role completeness is still only functionalism

The proposal is a physicalist, information-dynamical form of functionalism, but not unrestricted input-output functionalism. The roles require effective causal realization, a frozen system boundary, bounded temporal dynamics, and substrate-appropriate viability. A lookup table, behavioral imitation, or merely isomorphic diagram does not qualify. Critics may still deny that any causal-role identity captures phenomenality. The schematic paper does not settle that larger metaphysical dispute; it specifies the strongest intelligible version of the N^* claim within it.

Objection 8: The circuit analogy exaggerates the maturity of the theory

The analogy concerns a target epistemic state, not present evidential parity. Electrical component laws are mature; the N^* identity and its operationalizations are not. The research program succeeds only if training on its constructs eventually supports reliable readout, intervention, fault diagnosis, and transport across cases. Failure to achieve that competence would count against the claim of schematic legibility.

8. Empirical and Explanatory Payoffs

Schematic legibility adds demands beyond classification accuracy.

1. **Role-preserving transport.** If two substrates instantiate the same claimed phenomenal schematic, trained readers should identify the same role profile despite differences in material implementation. Superficial graph similarity is insufficient; causal organization and operating ranges must transport.
2. **Fault prediction.** Before an intervention, the model should specify which role is expected to fail and whether the result concerns absence, fragmentation, isolation, transience, altered character, or an unlicensed inference.
3. **Explanatory convergence.** Different valid estimators of one role should support the same schematic reading within tolerance. Persistent disagreement indicates that the relevant component is not yet readable.
4. **Compression without loss.** Experts should be able to move from detailed causal data to the role schematic and back to discriminating predictions. If the schematic discards every fact that distinguishes successful from failed cases, it is decorative rather than explanatory.
5. **Teachability.** Independent investigators should be able to learn and reproduce the readout rules on held-out cases. A conclusion available only to the theory's author is not schematic legibility.

These criteria turn a philosophical aspiration into a research standard. The model should not merely output consciousness. It should make investigators increasingly capable of seeing why a given organization receives that output, what would change it, and where the inference ceases to be licensed.

9. Conclusion

The familiar complaint against reductive materialism is that no inventory of physical parts would have led us to predict qualia. The complaint is compelling when the inventory is atomistic, externally described, and organizationally mute. Nothing in such a list identifies a bearer, binds a content into one state, makes that content system-wide, sustains it through a present interval, or differentiates its qualitative trajectory.

The N^* program proposes a different kind of physical description. It first identifies a candidate causal individual. It then asks whether a content is irreducibly integrated, counterfactually available across that individual, recurrently maintained, and actually realized in a viable regime. The character extension asks how the admitted content occupies and moves through a structured causal geometry. These are not independent signs placed near an occult phenomenal event. On the constitutive reading, they are the schematic organization of the event.

An electrical expert sees more than ink because a schematic has component semantics, operating conditions, compositional consequences, and a fault map. In the same way, a trained N^* reader should see more than a conjunction of scores. The reader should reconstruct a bearer, a unified system-level content, a bounded temporal presence, and a realized operating episode; predict what disappears when a role is removed; and know where evidence ends and the identity hypothesis begins.

The observer need not undergo the candidate's experience in order to observe that experience is occurring. What remains private is first-person acquaintance with the token episode, not necessarily the fact that the episode is instantiated. Asking for more would ask one experiencer to experience another's numerically distinct experience while both remain distinct experiencers. That is not a higher evidential standard. It is a contradiction in the specification of the test.

The resulting claim is neither that visualization proves consciousness nor that the hard problem has been dissolved by analogy. It is more precise:

> If $C_0 = V \wedge N^*$ is true, phenomenal presence is not hidden behind an otherwise complete dynamical description. It is structurally legible in that description. The theory's mature form would allow consciousness to be read from a realized organization as an engineer reads an operating state from a schematic.

That is the lacuna this paper fills. The core paper states the identity, the empirical papers discipline its application, and the schematic account explains what it would mean for the necessity to become intellectually visible rather than merely asserted.

References

Block, N. (1995). On a confusion about a function of consciousness. *Behavioral and Brain Sciences, 18*(2), 227–247. https://doi.org/10.1017/S0140525X00038188

Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies, 2*(3), 200–219.

Craver, C. F. (2007). *Explaining the brain: Mechanisms and the mosaic unity of neuroscience*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199299317.001.0001

Kripke, S. A. (1980). *Naming and necessity*. Harvard University Press.

Levine, J. (1983). Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly, 64*(4), 354–361. https://doi.org/10.1111/j.1468-0114.1983.tb00207.x

Machamer, P., Darden, L., & Craver, C. F. (2000). Thinking about mechanisms. *Philosophy of Science, 67*(1), 1–25. https://doi.org/10.1086/392759

Nagel, T. (1974). What is it like to be a bat? *The Philosophical Review, 83*(4), 435–450. https://doi.org/10.2307/2183914

Stilwell, P. (2026a). *C0 as N*: A minimal network-dynamics model of phenomenal consciousness* (Version 2.0) [Manuscript].

Stilwell, P. (2026b). *Indeterminacy as a scientific result: A four-outcome framework for consciousness attribution* [Manuscript].

Stilwell, P. (2026c). *Where is the conscious subject? A dynamical criterion for system boundaries* [Manuscript].

Stilwell, P. (2026d). *Consciousness without report: What system-wide availability actually requires* [Manuscript].

Stilwell, P. (2026e). *From phenomenal presence to phenomenal character: A dynamical-geometry extension of the N* model* [Manuscript].

Stilwell, P. (2026f). *Ablating N*: Does every conjunct earn its place?* [Manuscript].